

Does the Bond Market Need a Bank Account?

Forward-Measure Families, Pricing Kernels, and When the Savings Account Is a Numéraire

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1 Abstract

Practitioners price interest-rate derivatives with T -forward measures, each anchored by a traded bond; the textbook theory starts from a bank account that is not traded. We ask what the family of forward measures implies by itself. We give a definition of a consistent family that refers to no risk-neutral measure and show that it is equivalent to the existence of a pricing kernel, with no further regularity. Three consequences follow. First, pricing needs no bank account: the terminal bond is a numéraire. Second, for every finite tenor structure the roll-over account is a numéraire, so the discrete-time bank account always exists. Third, the continuous-time savings account need not be a numéraire: by a criterion of Döberlein and Schweizer (2001) it is one if and only if the terminal bond's generator factorises into a true martingale and a predictable process of finite variation, and an example that goes back to Platen's minimal market model, used for this purpose by Klein, Schmidt and Teichmann (2016), shows that this can fail with a smooth, decreasing term structure and zero short rate. The third point is known; what we add is the formulation that connects it with forward-measure practice, and an elementary treatment. Finally we explain why the question cannot be settled by data without a parametric model class, and relate it to the empirical literature on money-market bubbles.

2 1. Introduction

2.1 1.1 The Question

The replication argument of Black and Scholes (1973) and Merton (1973) prices a claim from the traded assets that hedge it. In a bond market the traded assets are zero-coupon bonds $p(\cdot, T)$. The bank account $B(t) = \exp(\int_0^t r(s) ds)$, the numéraire of the textbook theory (Harrison and Kreps 1979; Harrison and Pliska 1981), is not among them. It has to be synthesised by rolling over just-maturing bonds, a strategy that in continuous time trades through a continuum of maturities (Björk, Di Masi, Kabanov and Runggaldier 1997; Döberlein and Schweizer 2001). Practitioners do not need it: a caplet maturing at T is priced under the T -forward measure Q^T , with the T -bond as numéraire

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(Geman, El Karoui and Rochet 1995; Jamshidian 1997; Brace, Gatarek and Musiela 1997), and B never appears.

I met this discrepancy in my doctoral work (Wang 2008, Chapter 3), where the non-tradability of the bank account motivated a heuristic reformulation of the no-arbitrage hypothesis and a rigorous treatment was left open. This note takes the practitioner's objects as primitive and asks:

What does the family $\{Q^T\}$ of forward measures imply by itself? In particular, does it imply a bank account?

2.2 1.2 The Answer

The answer has three layers.

Pricing needs no bank account. We define a consistent family of forward measures without reference to a risk-neutral measure or an account (Definition 3.1) and show that such families are exactly those generated by a pricing kernel — a strictly positive adapted process ζ with $p(t, u) = E[\zeta_u \mid \mathcal{F}_t]/\zeta_t$ — and that the definition implies nothing more: no semimartingale property, no supermartingale property, no path regularity (Theorem 3.2, Corollary 3.3). The terminal bond is a numéraire and Q^{T^*} prices the market.

At every finite tenor structure the account exists and is a numéraire. For any finite set of roll-over dates, the roll-over strategy in just-maturing bonds, deflated by the terminal bond, is a true martingale under Q^{T^*} (Proposition 4.1). It defines a spot measure under which all bonds deflated by the roll-over account are martingales. This is the discrete-time bank account of Section 2 and the numéraire of the spot LIBOR measure of Jamshidian (1997); it is here that the true-martingale clause of the definition is used.

In continuous time the account need not be a numéraire. A savings account — a predictable numéraire of finite variation that induces the given family — exists if and only if the generator of the terminal bond factorises into a true martingale and a predictable process of finite variation, the property called goodness by Döberlein and Schweizer (2001) (Proposition 4.5). Consistency does not imply it. In an example that goes back to the minimal market model of Platen (Platen and Heath 2006) and was used for this purpose by Klein, Schmidt and Teichmann (2016), the term structure is smooth and strictly decreasing, the short rate is zero, the roll-overs converge to the constant account $B \equiv 1$ — and yet bond prices are below one and no savings account exists (Propositions 4.8 and 4.9, Appendix B). Every finite roll-over is priced at par by the terminal forward measure; the limit is not, and the mass that is lost is explicit (Remark 4.11).

Section 5 turns to data. The existence of a pricing kernel has no model-free testable content on a panel of bond prices, and whether the account is a numéraire is a statement about states a sample does not visit. Tests therefore need a parametric class; the existing evidence, from the literature on money-market bubbles, is of this kind. In this sense the bank account is a modelling convention.

2.3 1.3 What Is Ours and What Is Not

The third layer is known. That a savings account may fail to be a numéraire because a local martingale is strict is the core of the benchmark approach (Platen and Heath 2006) and of the literature on bubbles in the money-market account (Loewenstein and Willard 2000; Heston, Loewenstein and Willard 2007). Klein, Schmidt and Teichmann (2016; henceforth KST) develop the no-arbitrage theory of bond markets with the terminal bond as numéraire, state that a bank account that is not a numéraire is “not the exception but rather the rule”, and give the example we use in Section 4.3. The existence mechanism for savings accounts, goodness, is due to Döberlein and Schweizer (2001; henceforth D-S); uniqueness is due to Döberlein, Schweizer and Stricker (2000). The forward-measure approach to term structure modelling is due to Musiela and Rutkowski (1997).

What we add is: (i) a definition of forward-measure consistency that refers to no risk-neutral measure, and its identification with pricing kernels (Theorem 3.2) — the computation is standard, the identification is the point; (ii) the observation that every finite roll-over is a numéraire under the definition (Proposition 4.1), which is the precise sense in which the discrete-time bank account always exists; (iii) an exposition of the known negative result in the language of forward-measure families, with an elementary proof for the example; (iv) a discussion of identifiability. We claim no novelty for (iii) beyond the formulation.

3 2. Discrete Time: The Account as a Strategy

Let time be $t = 0, 1, \dots, N$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ with $\mathcal{F}_0$ trivial. The market consists of zero-coupon bonds $p(t, T) > 0$, $T = 1, \dots, N$, with $p(T, T) = 1$, and strictly positive risky assets $X^1, \dots, X^d$. No bank account is assumed. A bond that has matured is understood to be rolled into the account B of Proposition 2.1, so that all assets live on $\{0, \dots, N\}$. An arbitrage is a self-financing strategy with $V_0 = 0$, $V_N \geq 0$ a.s. and $\mathbb{P}(V_N > 0) > 0$.

Proposition 2.1 (Rolling account). *Define $B(0) := 1$ and $B(t) := \prod_{s=1}^t 1/p(s-1, s)$. Then B is the value process of the self-financing strategy that holds, during $(t-1, t]$, $B(t-1)/p(t-1, t)$ units of the bond maturing at t . Its one-period return $1/p(t-1, t) - 1$ is $\mathcal{F}_{t-1}$ -measurable: B is predictable.*

Proposition 2.2 (Bond-based FTAP in discrete time). *The following are equivalent:*

(NA) *there is no arbitrage;*

(EMM) *there is $Q \sim \mathbb{P}$ such that S/B is a Q -martingale for every traded asset S ;*

(RF) *there is $Q \sim \mathbb{P}$ such that, for every traded asset S , $E^Q[S(t)/S(t-1) - 1 \mid \mathcal{F}_{t-1}] = 1/p(t-1, t) - 1$;*

(BF) *for each $T \in \{1, \dots, N\}$ there is $Q^T \sim \mathbb{P}$ such that $S/p(\cdot, T)$ is a Q^T -martingale on $\{0, \dots, T\}$ for every traded asset S .*

Moreover, the forward measures obtained from Q in the proof of (EMM) $\Rightarrow$ (BF) satisfy, for $T_1 < T_2$,

$$\left. \frac{dQ^{T_1}}{dQ^{T_2}} \right|_{\mathcal{F}_{T_1}} = \frac{p(0, T_2)}{p(0, T_1) p(T_1, T_2)}, \quad (2.1)$$

a relation that involves bond prices only.

The proof assembles standard ingredients — the discrete-time fundamental theorem (Harrison and Pliska 1981 for finite Ω ; Dalang, Morton and Willinger 1990 in general), Bayes' formula, and the change of numéraire (Geman, El Karoui and Rochet 1995) — and is given in Appendix A. Two points matter for what follows. The account is a finite product of traded one-period returns: it exists as a strategy, and it is predictable, that is, locally riskless. And the forward measures derived from Q satisfy (2.1), in which neither Q nor B appears. Section 3 takes (2.1) as an axiom and drops Q and B .

4 3. Consistent Families of Forward Measures

4.1 3.1 Setup

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T^*]}, \mathbb{P})$ satisfy the usual conditions, with $\mathcal{F}_0$ trivial and $\mathcal{F} = \mathcal{F}_{T^*}$. For each maturity $T \in [0, T^*]$ the bond price $(p(t, T))_{t \in [0, T]}$ is strictly positive and adapted, with $p(T, T) = 1$. The market consists of the bonds only; currency is a unit of account and not a traded asset (further assets: Remark 3.6).

Martingales and local martingales are understood to have right-continuous paths with left limits (RCLL), and the bond ratios $p(\cdot, S)/p(\cdot, T)$ are assumed to have such paths. No path regularity and no semimartingale property is assumed for the nominal prices $p(\cdot, T)$ themselves; Theorem 3.2 will show that the definition below cannot deliver one.

4.2 3.2 Definition

Definition 3.1 (Consistent family of forward measures). A family $\{Q^T\}_{T \in [0, T^*]}$ of probability measures on $(\Omega, \mathcal{F})$, each equivalent to $\mathbb{P}$, is a *consistent family of forward measures* for the bond prices p if

- (FM) for all maturities S, T : $p(\cdot, S)/p(\cdot, T)$ is a Q^T -local martingale on $[0, S \wedge T]$;
- (TM) for every pair $T_1 < T_2$: $p(\cdot, T_1)/p(\cdot, T_2)$ is a *true* Q^{T_2} -martingale on $[0, T_1]$;
- (C) for every pair $T_1 < T_2$:

$$\left. \frac{dQ^{T_1}}{dQ^{T_2}} \right|_{\mathcal{F}_{T_1}} = \frac{p(0, T_2)}{p(0, T_1)p(T_1, T_2)}. \quad (3.1)$$

Only the restrictions $Q^T|_{\mathcal{F}_T}$ enter the definition; families with the same restrictions are identified.

Two comments. First, (3.1) is a genuine condition. Under Q^{T_2} , the change of numéraire from the T_2 -bond to the T_1 -bond produces a T_1 -forward measure with density (3.1); but in an incomplete market forward measures are not unique, and the family member Q^{T_1} need not be this one. Consistency is the requirement that the family's selections are mutually compatible. Second, (TM) is what makes the right-hand side of (3.1) a density: the bond ratio is a positive Q^{T_2} -local martingale with terminal value $1/p(T_1, T_2)$, and its expectation equals $p(0, T_1)/p(0, T_2)$ — so that (3.1) integrates to one — if and only if it is a true martingale. The definition thus *presupposes* the absence of bubbles in bond ratios (Cox and Hobson 2005; Jarrow, Protter and Shimbo 2010); it does not derive it.

4.3 3.3 The Characterization

Theorem 3.2 (Terminal coordinates). Let $Q^* \sim \mathbb{P}$ and let G be a strictly positive adapted process with

$$G_0 = 1, \quad G_{T^*} = \gamma \text{ (a deterministic constant)}, \quad \ell_u := E^{Q^*}[G_u] < \infty \quad (u \in [0, T^*]).$$

Let $L_t^u := E^{Q^*}[G_u | \mathcal{F}_t]$, $t \leq u$ (RCLL version), and define

$$p(t, u) := \frac{L_t^u}{G_t} \quad (0 \leq t \leq u \leq T^*), \quad \frac{dQ^u}{dQ^*} := \frac{G_u}{\ell_u} \text{ on } \mathcal{F}. \quad (3.2)$$

Then $\{Q^u\}$ is a consistent family of forward measures for p ; moreover $Q^{T^*} = Q^*$, $\gamma = p(0, T^*)$, and all bond ratios in (FM) are true martingales.

Conversely, if $\{Q^T\}$ is a consistent family for p , then $Q^* := Q^{T^*}$ and $G := p(0, T^*)/p(\cdot, T^*)$ satisfy the hypotheses, and (3.2) reproduces p and the restrictions $Q^T|_{\mathcal{F}_T}$.

Proof. ($\Leftarrow$) L^u is a strictly positive Q^* -martingale on $[0, u]$ with $L_u^u = G_u$ and $L_0^u = \ell_u$; its left limits are strictly positive as well. The density process of Q^u with respect to Q^* is $L_{t \wedge u}^u / \ell_u$, so $Q^u \sim Q^* \sim \mathbb{P}$. Clearly $p > 0$ and $p(u, u) = 1$. Since $G_{T^*} = \gamma$ is deterministic, $L^{T^*} \equiv \gamma$; hence $Q^{T^*} = Q^*$, $p(t, T^*) = \gamma / G_t$ and $p(0, T^*) = \gamma$. The ratios $p(\cdot, T)/p(\cdot, u) = L^T / L^u$ are RCLL.

Fix maturities T, u and $s \leq t \leq T \wedge u$. Bayes' formula with density process L^u / ℓ_u gives

$$E^{Q^u} \left[\frac{p(t, T)}{p(t, u)} \mid \mathcal{F}_s \right] = \frac{E^{Q^*} \left[\frac{L_t^u}{\ell_u} \cdot \frac{L_t^T}{L_t^u} \mid \mathcal{F}_s \right]}{L_s^u / \ell_u} = \frac{L_s^T}{L_s^u} = \frac{p(s, T)}{p(s, u)},$$

which is valid for nonnegative integrands; $s = 0$ gives $E^{Q^u}[p(t, T)/p(t, u)] = \ell_T / \ell_u < \infty$. This proves (FM) and (TM), with true martingales throughout. For (C), let $T_1 < T_2$; then

$$\left. \frac{dQ^{T_1}}{dQ^{T_2}} \right|_{\mathcal{F}_{T_1}} = \frac{G_{T_1} / \ell_{T_1}}{L_{T_1}^{T_2} / \ell_{T_2}} = \frac{\ell_{T_2}}{\ell_{T_1}} \cdot \frac{1}{p(T_1, T_2)},$$

and $\ell_{T_i} = L_0^{T_i} / G_0 = p(0, T_i)$.

($\Rightarrow$) G is strictly positive and adapted, $G_0 = 1$, and $G_{T^*} = p(0, T^*)$ is deterministic. By (TM) for the pair (u, T^*) , $E^{Q^*}[1/p(u, T^*) | \mathcal{F}_t] = p(t, u)/p(t, T^*)$, that is, $E^{Q^*}[G_u | \mathcal{F}_t] = G_t p(t, u)$; with $t = 0$, $\ell_u = p(0, u) < \infty$. This is the first formula in (3.2). Let Λ^u be the density process of Q^u with respect to Q^* . By (C) for (u, T^*) , $\Lambda_u^u = p(0, T^*)/(p(0, u)p(u, T^*)) = G_u / \ell_u$, and since Λ^u is a Q^* -martingale, $\Lambda_t^u = L_t^u / \ell_u$ for $t \leq u$. $\square$

Corollary 3.3. (a) (Against-terminal consistency suffices.) Definition 3.1 is equivalent to its restriction to the pairs (u, T^*) : conditions (TM) and (C) for (u, T^*) , $u \in [0, T^*]$.

(b) (Kernel coordinates.) Let ζ be a strictly positive adapted process with $\zeta_0 = 1$ and $E[\zeta_u] < \infty$ for all u . Then

$$p(t, u) := \frac{E[\zeta_u | \mathcal{F}_t]}{\zeta_t}, \quad \frac{dQ^u}{d\mathbb{P}} := \frac{\zeta_u}{E[\zeta_u]} \text{ on } \mathcal{F} \quad (3.3)$$

define a consistent family. Its terminal pair is $Q^* = Q^{T^*}$ and $G = \zeta/Z^*$, where $Z_t^* = E[\zeta_{T^*} \mid \mathcal{F}_t]/E[\zeta_{T^*}]$ is the $\mathbb{P}$ -density process of Q^* . Conversely, every consistent family is generated in this way by the unique process

$$\zeta_u = p(0, u) \frac{dQ^u}{d\mathbb{P}} \Big|_{\mathcal{F}_u} = Z_u^* G_u.$$

Proof. (a) The direction ($\Rightarrow$) of Theorem 3.2 uses only (TM) and (C) for the pairs (u, T^*) ; the direction ($\Leftarrow$) returns the full definition. (b) The computation in the proof of Theorem 3.2 ($\Leftarrow$) uses that G_{T^*} is deterministic only to conclude $Q^{T^*} = Q^*$; everything else goes through verbatim for $(\mathbb{P}, \zeta)$ in place of (Q^*, G) . For the terminal pair, $G = p(0, T^*)/p(\cdot, T^*) = E[\zeta_{T^*}]/E[\zeta_{T^*} \mid \mathcal{F}_\cdot] = \zeta/Z^*$. Conversely, with (Q^*, G) from Theorem 3.2 put $\zeta := Z^* G$; then $\frac{dQ^u}{d\mathbb{P}} \Big|_{\mathcal{F}_u} = \frac{G_u}{\ell_u} Z_u^* = \zeta_u/p(0, u)$ and $E[\zeta_u \mid \mathcal{F}_t]/\zeta_t = E^{Q^*}[G_u \mid \mathcal{F}_t]/G_t = p(t, u)$. $\square$

In words: a consistent family of forward measures is the same thing as a pricing kernel. The terminal normalisation $G_{T^*} = \text{const}$ is a convention that singles out the representative $(Q^{T^*}, p(0, T^*)/p(\cdot, T^*))$.

4.4 3.4 Remarks

Remark 3.4 (No regularity is implied). By Theorem 3.2, Definition 3.1 implies nothing about G beyond adaptedness, positivity and integrability at fixed times. In particular it cannot deliver a semi-martingale property for nominal bond prices. One regularity property has a simple meaning: $p(t, u) \leq 1$ for all $t \leq u$ if and only if G is a Q^* -supermartingale, if and only if ζ is a $\mathbb{P}$ -supermartingale ($E^{Q^*}[G_u \mid \mathcal{F}_t] = G_t p(t, u)$). This is the setting of the potential approach (Rogers 1997).

Remark 3.5 (Positioning). ζ is a state-price density: Constantinides (1992), Flesaker and Hughston (1996), Rogers (1997), Jin and Glasserman (2001). Defining T -forward measures from a kernel through the martingale $\zeta_t p(t, T)/p(0, T)$ is standard (for instance Qin and Linetsky 2017, Section 2). The computation behind Theorem 3.2 is therefore not new; its content is the identification. Definition 3.1 — an axiomatisation in the spirit of the forward-measure approach of Musiela and Rutkowski (1997) — is equivalent to the existence of a pricing kernel and implies nothing beyond it. In the language of D–S the bond prices are “generated by (Q^*, G) ”, with every regularity requirement on the generator removed. Finally, for a consistent family Q^{T^*} is an equivalent measure under which all terminal-bond-deflated bond prices are martingales; under the regularity assumptions of KST this implies their condition of no asymptotic free lunch (their Theorem 5.2). Conversely, no asymptotic free lunch yields local martingales only. The difference between the two frameworks is the true-martingale clause (TM).

Remark 3.6 (Further traded assets). If the market contains further assets S , and (FM) is required for them as well, Theorem 3.2 holds with the additional condition that $S/p(\cdot, T^*)$ is a Q^* -local martingale for each S : by Bayes’ formula, $\frac{S_t}{p(t, u)} \cdot \frac{L_t^u}{\ell_u} = \frac{S_t G_t}{\ell_u} = \frac{\gamma}{\ell_u} \cdot \frac{S_t}{p(t, T^*)}$. This is a condition on the pair (S, G) . For instance, if currency were storable and tradable in both directions ($S \equiv 1$), G would have to be a Q^* -local martingale.

5 4. The Bank Account

5.1 4.1 Finite Roll-Overs

Let $\pi : 0 = t_0 < t_1 < \dots < t_k = T^*$ be a finite set of roll-over dates. The roll-over strategy starts with one unit of currency, buys bonds maturing at t_1 , and at each t_j reinvests the proceeds in bonds maturing at t_{j+1} . With $\beta_j := \prod_{i=1}^j 1/p(t_{i-1}, t_i)$ (so $\beta_0 = 1$, and β_j is $\mathcal{F}_{t_{j-1}}$ -measurable), its value deflated by the terminal bond is

$$V_0^\pi := \frac{1}{p(0, T^*)}, \quad V_t^\pi := \beta_j \frac{p(t, t_j)}{p(t, T^*)}, \quad t \in (t_{j-1}, t_j],$$

and its nominal value is $B_t^\pi := V_t^\pi p(t, T^*)$; in particular $B_{t_j}^\pi = \beta_j$ and $B_{T^*}^\pi = V_{T^*}^\pi$.

Proposition 4.1 (Every finite roll-over is a numéraire). *Let $\{Q^T\}$ be a consistent family and $Q^* = Q^{T^*}$. Then V^π is a strictly positive true Q^* -martingale on $[0, T^*]$. Consequently:*

(a) $p(0, T^*) E^{Q^*} [B_{T^*}^\pi] = 1$: the terminal forward measure prices the roll-over at par;

(b) $dQ^\pi/dQ^* := p(0, T^*) B_{T^*}^\pi$ defines a probability measure $Q^\pi \sim \mathbb{P}$ (the spot measure of π), and each $p(\cdot, u)/B^\pi$ is a true Q^π -martingale on $[0, u]$;

(c) (Q^π, B^π) induces the given family: $dQ^T/dQ^\pi|_{\mathcal{F}_t} = p(t, T)/(B_t^\pi p(0, T))$ for $t \leq T$.

Only (TM) for the pairs (t_j, T^*) is used.

Proof. On $[t_{j-1}, t_j]$ we have $V^\pi = \beta_j R^j$, where $R^j := p(\cdot, t_j)/p(\cdot, T^*)$ is a true Q^* -martingale on $[0, t_j]$ by (TM) and $\beta_j > 0$ is $\mathcal{F}_{t_{j-1}}$ -measurable. The pieces fit together: $\beta_j R_{t_{j-1}}^j = \beta_{j-1}/p(t_{j-1}, T^*) = V_{t_{j-1}}^\pi$. For $t_{j-1} \leq s \leq t \leq t_j$, conditional expectations of nonnegative variables give $E^{Q^*} [V_t^\pi | \mathcal{F}_s] = \beta_j R_s^j = V_s^\pi$. Taking expectations and inducting over j , $E^{Q^*} [V_t^\pi] = V_0^\pi = 1/p(0, T^*) < \infty$ for all t , so V^π is integrable, and the tower property yields the martingale property on $[0, T^*]$. This gives (a) and the first half of (b). The rest is Bayes' formula: the Q^* -density process of Q^π is $p(0, T^*)V^\pi$, and

$$\frac{p(t, u)}{B_t^\pi} \cdot p(0, T^*) V_t^\pi = p(0, T^*) \frac{p(t, u)}{p(t, T^*)}, \quad \frac{L_t^T/\ell_T}{p(0, T^*) V_t^\pi} = \frac{p(t, T)}{B_t^\pi p(0, T)}. \quad \square$$

Remark 4.2. With the tenor dates of a LIBOR market model, B^π is the discretely rebalanced account and Q^π the spot LIBOR measure of Jamshidian (1997). With discrete trading dates, B^π is the rolling account of Proposition 2.1. Proposition 4.1 says that under Definition 3.1 such an account always exists as a numéraire; no condition beyond consistency is needed. Between roll-over dates B^π carries the price risk of the bond currently held: it is locally riskless only on the grid.

5.2 4.2 The Savings Account in Continuous Time

Let $\mathcal{A}_1^+$ denote the class of strictly positive, predictable, RCLL processes C of finite variation with $C_0 = 1$ and strictly positive left limits; then $C \in \mathcal{A}_1^+$ if and only if $1/C \in \mathcal{A}_1^+$.

Definition 4.3 (Implied savings account). An *implied savings account* for the family $\{Q^T\}$ is a pair (Q, B) with $Q \sim \mathbb{P}$ on $\mathcal{F}$ and $B \in \mathcal{A}_1^+$ such that, for all $t \leq T \leq T^*$,

$$p(t, T) = E^Q \left[\frac{B_t}{B_T} \mid \mathcal{F}_t \right] \quad \text{and} \quad \frac{dQ^T}{dQ} \Big|_{\mathcal{F}_t} = \frac{p(t, T)}{B_t p(0, T)}.$$

An *implied savings account for the prices* is a pair $(\tilde{Q}, \tilde{B})$ that satisfies the first condition alone.

The second notion is that of Musiela and Rutkowski (1997), Döberlein, Schweizer and Stricker (2000) and D-S; the first adds the requirement that the account induce the given family. The account is predictable and of finite variation — locally riskless — which is what distinguishes it from the terminal bond and from the finite roll-overs.

Definition 4.4 (Goodness; after D-S). The terminal pair (Q^*, G) is *good* if $G = MC$ with M a strictly positive true Q^* -martingale, $M_0 = 1$, and $C \in \mathcal{A}_1^+$.

Proposition 4.5 (After D-S). For a consistent family with terminal pair (Q^*, G) and kernel ζ the following are equivalent:

- (i) an implied savings account for the family exists;
- (ii) (Q^*, G) is good;
- (iii) $\zeta = NC$ with N a strictly positive true $\mathbb{P}$ -martingale, $N_0 = 1$, and $C \in \mathcal{A}_1^+$.

In that case $B = 1/C$, $dQ/dQ^* = M_{T^*}$ and $dQ/d\mathbb{P} = N_{T^*}$.

Proof. (ii) $\Rightarrow$ (i). Put $dQ/dQ^*|_{\mathcal{F}_t} := M_t$ and $B := 1/C$; since M is a strictly positive true martingale with $M_0 = 1$, Q is a probability measure equivalent to $\mathbb{P}$, and $B \in \mathcal{A}_1^+$. By Bayes' formula, $p(t, T) = E^{Q^*}[M_T C_T \mid \mathcal{F}_t]/(M_t C_t) = E^Q[C_T \mid \mathcal{F}_t]/C_t = E^Q[B_t/B_T \mid \mathcal{F}_t]$. The T -forward measure induced by (Q, B) has Q^* -density process $M_t p(t, T)/(B_t p(0, T)) = G_t p(t, T)/p(0, T) = L_t^T/\ell_T$, which is the density process of Q^T .

(i) $\Rightarrow$ (ii). The second condition of Definition 4.3 at $T = T^*$ gives $dQ^*/dQ|_{\mathcal{F}_t} = p(t, T^*)/(B_t p(0, T^*)) = 1/(B_t G_t)$. Hence $M := BG$ is the Q^* -density process of Q , a strictly positive true Q^* -martingale with $M_0 = 1$, and $G = M \cdot (1/B)$ with $1/B \in \mathcal{A}_1^+$.

(ii) $\Leftrightarrow$ (iii). Since $\zeta = Z^*G$, we have $G = MC$ if and only if $\zeta = NC$ with $N = Z^*M$; and by Bayes' formula N is a true $\mathbb{P}$ -martingale if and only if M is a true Q^* -martingale. $\square$

The mechanism — a good pair yields the account $B = 1/C$ — is Theorem 5 of D-S. Proposition 4.5 restates it for families and adds the converse, which the coincidence requirement makes immediate.

Remark 4.6 (Uniqueness, the candidate account, and the defect). A product MC of a local martingale and a predictable process of finite variation is a special semimartingale (Jacod and Shiryaev 2003, Chapter I), and the multiplicative decomposition of a special semimartingale with strictly positive left limits is unique (Jacod 1979). D-S state goodness in this form: G is special and the local-martingale factor of its multiplicative decomposition is a true martingale. Uniqueness has two consequences. The implied savings account is unique (Döberlein, Schweizer and Stricker 2000). And if G is special with

decomposition $G = MC$, there is exactly one *candidate* account, $B = 1/C$. Since M is a positive local martingale, hence a supermartingale, and $M_{T^*} = p(0, T^*) B_{T^*}$,

$$p(0, T^*) E^{Q^*} [B_{T^*}] \leq 1, \quad \text{with equality if and only if the pair is good.}$$

The terminal forward measure prices the terminal value of the candidate account at or below par; goodness is pricing at par. The defect $\delta := 1 - p(0, T^*) E^{Q^*} [B_{T^*}]$ is the mass lost by the candidate risk-neutral density M_{T^*} .

Remark 4.7 (Which failures can occur). If rates are nonnegative ($p \leq 1$) and $u \mapsto p(0, u)$ is right-continuous, then G , in an RCLL modification, is a strictly positive supermartingale and hence a special semimartingale with strictly positive left limits. Goodness can then fail only because M is a strict local martingale, as in Section 4.3. Without a sign restriction G need not be a semimartingale at all: KST (Remark 8.4) generate bond prices from the account $\exp(Z)$ with Z a fractional Brownian motion. In the bond-only market this creates no free lunch, because terminal-bond-deflated prices remain true martingales. The semimartingale theorem of Delbaen and Schachermayer (1994) concerns prices in units of a traded numéraire, which currency is not. A generator that is not a semimartingale obstructs the savings account, not the absence of arbitrage.

5.3 4.3 An Example in Which the Account Is Not a Numéraire

The following model is the case $A \equiv 1$ of KST (Introduction, Example 2.12 and Section 8.1), who take it from the minimal market model of the benchmark approach (Platen and Heath 2006).

Let W be a four-dimensional Brownian motion, $(\mathcal{F}_t)$ its augmented natural filtration, $x \in \mathbb{R}^4 \setminus \{0\}$, $x_0 := \|x\|^2$, and

$$X_t := \|x + W_t\|^2, \quad \zeta_t := \frac{x_0}{X_t}, \quad t \in [0, T^*].$$

Proposition 4.8 (The model). (a) X is a squared Bessel process of dimension four, $dX_t = 4 dt + 2\sqrt{X_t} d\beta_t$, with the Brownian motion $\beta_t := \int_0^t \langle n_s, dW_s \rangle$, $n_s := (x + W_s)/\|x + W_s\|$; $X_t > 0$ for all t , a.s.; and ζ is a continuous, strictly positive $\mathbb{P}$ -local martingale with $d\zeta_t = -\theta_t \zeta_t d\beta_t$, $\theta_t := 2/\sqrt{X_t}$.

(b) For $t < u$, $E[\zeta_u | \mathcal{F}_t] = \zeta_t (1 - \exp(-X_t/(2(u-t))))$. In particular $E[\zeta_u] = 1 - e^{-x_0/(2u)} < 1$ for every $u > 0$: ζ is a strict local martingale on every interval $[0, u]$.

(c) The family generated by ζ through (3.3) is consistent, with

$$p(t, u) = 1 - \exp\left(-\frac{X_t}{2(u-t)}\right), \quad t < u \leq T^*. \quad (4.1)$$

The map $(t, u) \mapsto p(t, u)$ is jointly continuous on $\{t \leq u\}$ with values in $(0, 1]$, and smooth and strictly decreasing in $u > t$. The forward rates $f(t, u) = -\partial_u \log p(t, u)$ are strictly positive, and $f(t, u) \rightarrow 0$ as $u \downarrow t$: the short rate is zero.

(d) The terminal generator $G = p(0, T^*)/p(\cdot, T^*)$ is a continuous Q^* -local martingale and is strict on every horizon: $E^{Q^*} [G_u] = p(0, u) < 1$ for $u > 0$.

Proof. (a) Itô's formula and Lévy's characterisation; $x + W$ does not hit the origin in dimension four (Revuz and Yor 1999, Chapter XI), and $d(1/X) = -X^{-2}dX + X^{-3}d\langle X \rangle = -2X^{-3/2}d\beta$.

- (b) By the Markov property it suffices to show that $E[\|y + W_s\|^{-2}] = (1 - e^{-\|y\|^2/(2s)})/\|y\|^2$ for $y \neq 0$. The variable $\|y + W_s\|^2/s$ is noncentral χ^2 with four degrees of freedom and noncentrality $\lambda = \|y\|^2/s$, that is, a Poisson($\lambda/2$)-mixture of central χ_{4+2j}^2 ; since $E[1/\chi_k^2] = 1/(k-2)$ for $k > 2$,

$$E\left[\frac{s}{\|y + W_s\|^2}\right] = \sum_{j \geq 0} e^{-\lambda/2} \frac{(\lambda/2)^j}{j!} \cdot \frac{1}{2j+2} = \frac{1 - e^{-\lambda/2}}{\lambda}.$$

- (c) Corollary 3.3(b). Joint continuity at the diagonal holds because X is continuous and strictly positive, so that $X_t/(2(u-t)) \rightarrow \infty$ as $u-t \downarrow 0$. With $a := X_t/2$ and $s := u-t$, $f(t, u) = (a/s^2)/(e^{a/s} - 1)$.
- (d) By Corollary 3.3(b), $Z_t^* = \zeta_t p(t, T^*)/p(0, T^*)$ and $GZ^* = \zeta$, a continuous $\mathbb{P}$ -local martingale; hence G is a continuous Q^* -local martingale, and $E^{Q^*}[G_u] = E[Z_u^* G_u] = E[\zeta_u] = p(0, u)$. $\square$

Proposition 4.9 (No savings account for the family). *In the model of Proposition 4.8 no implied savings account for the family exists. Equivalently, the terminal pair is not good.*

Proof. Suppose (Q, B) is an implied savings account for the family. As in the proof of Proposition 4.5, $M := BG$ is the Q^* -density process of Q : a strictly positive true Q^* -martingale with $M_0 = 1$. The filtration is Brownian and $Q^* \sim \mathbb{P}$, so every Q^* -martingale is continuous (MZ^* is a continuous $\mathbb{P}$ -martingale and $Z^* > 0$ is continuous). Hence M is continuous, and so is $B = M/G$, which is of finite variation by definition. Integration by parts, with $[B, G] = 0$, gives $M = 1 + \int B dG + \int G dB$. Therefore $\int G dB = M - 1 - \int B dG$ is a continuous local martingale of finite variation that starts at zero, hence it vanishes identically. Since $G > 0$, $B \equiv 1$ and $M = G$. But $E^{Q^*}[M_{T^*}] = 1$, whereas $G_{T^*} = p(0, T^*) < 1$ is a constant: a contradiction. The equivalence with goodness is Proposition 4.5. $\square$

The argument uses Bayes' formula, the continuity of martingales in a Brownian filtration, the product rule, and the fact that a continuous local martingale of finite variation is constant. In the terms of Remark 4.6, G is its own local-martingale factor, the candidate account is $B \equiv 1$, and the defect is $\delta = 1 - p(0, T^*) = e^{-x_0/(2T^*)}$. Appendix B shows more: the prices (4.1) admit no implied savings account at all, whatever the measure.

Remark 4.10 (Attribution). KST show for this model that a T^* -forward measure exists under which all bond ratios are true martingales, that the roll-overs converge to the account $B \equiv 1$ (their Lemma 8.1), and that $B/p(\cdot, T^*)$ is a strict Q^* -local martingale, so that B is not a numéraire. Added here are the observation that the model is a consistent family in the sense of Definition 3.1 (by Corollary 3.3), the statement in terms of savings accounts and goodness (Proposition 4.9), and Appendix B. The exponent printed in their formula (8.2) is $\|x + W_t\|^{-2}$; the computation in Proposition 4.8(b) gives $\|x + W_t\|^2$, as in the bond-price formula of the minimal market model. Their Lemma 8.1 is not affected (see Remark 4.11).

Remark 4.11 (What happens). The short rate is zero and the limit of the roll-overs is currency held

over time, $B \equiv 1$; nevertheless $p(0, T^*) < 1$. One unit of currency at T^* costs $p(0, T^*)$ through the bond and 1 through the account — the “less expensive” long-dated bond of the benchmark approach. If currency is added as a traded asset, buying the bond and shorting the account is an arbitrage in units of the account; in units of the terminal bond the short position, $-G/p(0, T^*)$, is unbounded below and not admissible. This is the dependence of admissibility on the numéraire described by Delbaen and Schachermayer (1995); in the terminology of KST the account is in a bubble state. Every *individual* bond deflated by the terminal bond is a true Q^* -martingale; the self-financing roll-over is one for every finite set of dates (Proposition 4.1); only the limit fails. Precisely: for partitions π_n with k_n dates and mesh at most KT^*/k_n ,

$$1 \leq B_{T^*}^{\pi_n} = \prod_{i=1}^{k_n} \left(1 - e^{-X_{t_{i-1}}/(2(t_i - t_{i-1}))}\right)^{-1} \leq \left(1 - e^{-mk_n/(2KT^*)}\right)^{-k_n} \rightarrow 1 \quad \text{a.s.},$$

where $m := \min_{s \leq T^*} X_s > 0$ (this is the argument of KST, Lemma 8.1). Yet $p(0, T^*) E^{Q^*}[B_{T^*}^{\pi_n}] = 1$ for every n . The densities of the spot measures Q^{π_n} converge almost surely to the constant $p(0, T^*) < 1$ although each has mean one: the roll-over payoffs are not uniformly integrable under the terminal forward measure, and the mass $e^{-x_0/(2T^*)}$ is lost in the limit. KST interpret this loss as an illiquidity premium and relate it to multiple yield curves.

6 5. What Data Can and Cannot Say

Two hypotheses are nested in the title question.

(H0) The bond market admits a pricing kernel — equivalently (Theorem 3.2), a consistent family of forward measures exists.

(H1) The kernel factorises as in Proposition 4.5(iii) — equivalently, the savings account is a numéraire.

H0 has no model-free content on a panel of zero-coupon prices. H0 restricts the law of the price processes, and a sample is one path. Apart from static bounds (positivity; with storable currency, monotonicity in maturity), no finite panel can contradict it without further assumptions. Every sharper implication is a joint implication with a parametric specification of the real-world dynamics and of the market price of risk, and a rejection is a rejection of the specification. The best-known instance is the rejection of completely affine prices of risk (Duffee 2002; Dai and Singleton 2002). For Gaussian models the no-arbitrage restrictions are, moreover, nearly irrelevant for the real-world dynamics of yields (Joslin, Singleton and Zhu 2011; Duffee 2011). Segment-wise calibrations of one model class — such as the exercise documented in the repository that accompanies this note — belong here: disagreement of the calibrated prices of risk across maturity segments is a diagnostic of the class, not evidence on H0.

H1 is a statement about states the sample does not visit. Whether a positive local martingale is a true martingale is decided by its behaviour at extreme levels; for one-dimensional diffusions, by the growth of the volatility function (Delbaen and Shirakawa 2002; Mijatović and Urusov 2012). When the factor is

strict, the candidate risk-neutral measure still exists as a measure but is not equivalent to $\mathbb{P}$: it charges an explosion event that is $\mathbb{P}$ -null (Föllmer’s exit measure; see Carr, Fisher and Ruf 2014). In parametric examples this is visible directly: a volatility factor that stays positive under $\mathbb{P}$ reaches zero under the candidate measure (Gnoatto, Grasselli and Platen 2018, Section 3.3). Heuristically, a model in which the account is not a numéraire is the limit of models in which it is, each containing a state of vanishing real-world probability and non-vanishing risk-neutral weight — a peso problem carried to its limit — and on a sample in which that state has not occurred the two cannot be told apart. A test of H1 must therefore extrapolate: non-parametrically, the volatility function beyond the range of the data (Jarrow, Kchia and Protter 2011); parametrically, through the region of the parameter space in which the factor is strict.

The existing evidence is parametric. Heston, Loewenstein and Willard (2007) show that a bubble in the money-market account exists if and only if no equivalent risk-neutral measure exists. Baldeaux, Ignatieva and Platen (2018) estimate several stochastic-volatility models for a proxy of the numéraire portfolio by Markov chain Monte Carlo and find the strict region in all of them. Gnoatto, Grasselli and Platen (2018) calibrate a multi-currency model to FX option surfaces; the condition that separates the two regimes is a parameter inequality whose calibrated value lies close to zero and changes sign across currencies and calibration dates, and where it fails, long-dated zero-coupon bonds are substantially cheaper than their formal risk-neutral values. These are statements within model classes. Whether the savings account of an actual bond market is a numéraire is, in our reading, not a question that prices can settle without such a class.

7 6. Conclusion

Does the bond market need a bank account? For pricing, no: a consistent family of forward measures is a pricing kernel in other coordinates, the terminal bond is a numéraire, and nothing else is implied. At every finite tenor structure the market has an account anyway: the roll-over is a numéraire, with its own spot measure, under consistency alone. What consistency does not deliver is the continuous-time savings account as a numéraire. The criterion is the goodness of Döberlein and Schweizer; the example of Platen and of Klein, Schmidt and Teichmann shows that it can fail in a model with a smooth term structure and zero short rate, where every finite roll-over is priced at par and the limit is not.

Term structure models that start from a short rate or from a savings account — short-rate models, the Heath–Jarrow–Morton framework under its drift condition — impose goodness by construction; D–S give sufficient conditions and examples. Models that start from a numéraire portfolio, as the benchmark approach does, do not impose it, and their less expensive long-dated bonds are the non-good case. Section 5 argues that prices alone cannot decide between the two without a parametric class. The bank account of continuous-time theory is, in this sense, a modelling convention: harmless at every finite tenor structure, and an assumption in the limit.

8 Acknowledgments

This note grew out of a puzzle I met in my doctoral work (Wang 2008, Chapter 3): the standard theory relies on a bank account that is not traded. It was developed in extended collaboration with Claude (Anthropic), which contributed to the formulation of Definition 3.1, located the relevant literature — in particular the example of Section 4.3 in Klein, Schmidt and Teichmann — drafted proofs, and ran adversarial reviews of earlier versions. One of these reviews showed that an example I had intended to use was degenerate and that the phenomenon was already known; the note was rewritten accordingly. I have checked the arguments presented here; all errors are mine. Code for the calibration exercise mentioned in Section 5 is available at <https://github.com/shaohui1977/bond-family-na>.

9 Appendix A: Proofs for Section 2

Proof of Proposition 2.1. Holding $B(t-1)/p(t-1, t)$ units of the bond maturing at t costs $B(t-1)$ at time $t-1$ and pays $B(t-1)/p(t-1, t) = B(t)$ at time t ; the strategy is self-financing, and $B(t)$ is $\mathcal{F}_{t-1}$ -measurable. $\square$

Lemma A.1 (Adding the account changes nothing). *The terminal payoffs attainable by self-financing strategies in the original market coincide with those attainable when B is added as an asset.*

Proof. A position in B is replicated by the rolling strategy of Proposition 2.1, which is self-financing in the original market. $\square$

Lemma A.2 (Numéraire invariance). *Let N be a strictly positive traded asset or self-financing portfolio, let Q be a measure under which S/N is a Q -martingale for every traded asset S , and let V be the value of a self-financing strategy with bounded holdings. Then V/N is a Q -martingale.*

Proof. Self-financing gives $V_t/N_t - V_{t-1}/N_{t-1} = \sum_j \theta_t^j \Delta(S^j/N)_t$ with θ_t predictable and bounded; each term is a Q -martingale difference. (For general holdings V/N is a generalised martingale, which suffices below because $V_N \geq 0$.) $\square$

Proof of Proposition 2.2. (NA) $\Leftrightarrow$ (EMM). By Proposition 2.1, B is a strictly positive self-financing portfolio, and by Lemma A.1 adding it as an asset does not change the attainable payoffs. The discrete-time fundamental theorem of asset pricing (Dalang, Morton and Willinger 1990; Harrison and Pliska 1981 for finite Ω), applied with B as numéraire, gives the equivalence.

(EMM) $\Leftrightarrow$ (RF). For a traded asset $S > 0$, $E^Q[S(t)/B(t) \mid \mathcal{F}_{t-1}] = S(t-1)/B(t-1)$ if and only if $E^Q[S(t)/S(t-1) \mid \mathcal{F}_{t-1}] = B(t)/B(t-1) = 1/p(t-1, t)$; all steps multiply by strictly positive $\mathcal{F}_{t-1}$ -measurable quantities.

(EMM) $\Rightarrow$ (BF). For each T put $Z_t^T := p(t, T)/(B(t)p(0, T))$, $t \leq T$; it is a strictly positive Q -martingale with $Z_0^T = 1$ and defines $Q^T \sim \mathbb{P}$ by $dQ^T/dQ := Z_T^T$. By Bayes' formula, for $s \leq t \leq T$,

$$E^{Q^T} \left[\frac{S(t)}{p(t, T)} \mid \mathcal{F}_s \right] = \frac{E^Q [Z_t^T S(t)/p(t, T) \mid \mathcal{F}_s]}{Z_s^T} = \frac{E^Q [S(t)/B(t) \mid \mathcal{F}_s] B(s)}{p(s, T)} = \frac{S(s)}{p(s, T)}.$$

(BF) $\Rightarrow$ (NA). Let $V_0 = 0$ and $V_N \geq 0$ a.s. By (BF) with $T = N$ and Lemma A.2, $V/p(\cdot, N)$ is a (generalised) Q^N -martingale, so $0 = V_0/p(0, N) = E^{Q^N}[V_N]$ and $V_N = 0$ a.s.

Relation (2.1). Under (EMM), $\frac{dQ^{T_1}}{dQ^{T_2}}|_{\mathcal{F}_{T_1}} = Z_{T_1}^{T_1}/Z_{T_1}^{T_2} = \frac{p(0, T_2)}{p(0, T_1)p(T_1, T_2)}$; the factor $B(T_1)$ cancels. $\square$

10 Appendix B: No Savings Account for the Prices of the Example

Proposition B.1. *For the prices (4.1) there is no pair $(\tilde{Q}, \tilde{B})$ with $\tilde{Q} \sim \mathbb{P}$ on $\mathcal{F}_{T^*}$, $\tilde{B} \in \mathcal{A}_1^+$ and $p(t, T) = E^{\tilde{Q}}[\tilde{B}_t/\tilde{B}_T | \mathcal{F}_t]$ for all $t \leq T \leq T^*$.*

Proof. Suppose $(\tilde{Q}, \tilde{B})$ is such a pair and let $\tilde{Z}$ be the $\mathbb{P}$ -density process of $\tilde{Q}$. Since $(\mathcal{F}_t)$ is the augmented filtration of W , the martingale representation theorem gives $\tilde{Z} = \mathcal{E}(-\int \psi \cdot dW)$ for a predictable $\mathbb{R}^4$ -valued ψ with $\int_0^{T^*} |\psi_s|^2 ds < \infty$ a.s.; by Girsanov's theorem $\tilde{W} := W + \int \psi ds$ is a $\tilde{Q}$ -Brownian motion (Karatzas and Shreve 1991, Sections 3.4–3.5); and every $\tilde{Q}$ -local martingale is continuous.

Step 1 ($\tilde{B}$ is continuous). Fix T . $N_t^T := E^{\tilde{Q}}[1/\tilde{B}_T | \mathcal{F}_t]$ is a strictly positive continuous $\tilde{Q}$ -martingale, and $p(t, T) = \tilde{B}_t N_t^T$ for each t , hence, by right-continuity, for all $t \leq T$ simultaneously. Since $p(\cdot, T)$ and N^T are continuous, so is $\tilde{B}$ on $[0, T]$; take $T = T^*$.

Step 2 (bond dynamics). We have $p(t, T) = f(T - t, X_t)$ with $f(s, x) := 1 - e^{-x/(2s)}$. On $[0, T]$ Itô's formula gives

$$dp(t, T) = p(t, T) v(t, T) (d\beta_t + \theta_t dt), \quad v(t, T) := \frac{2}{\sqrt{X_t}} \cdot \frac{y}{e^y - 1}, \quad y := \frac{X_t}{2(T - t)},$$

because $2\sqrt{x} \partial_x f = f v$ and $-\partial_s f + 4 \partial_x f + 2x \partial_{xx} f = 4 \partial_x f = f v \theta$. Under $\tilde{Q}$, $\tilde{\beta} := \int \langle n_s, d\tilde{W}_s \rangle = \beta + \int \langle n_s, \psi_s \rangle ds$ is a Brownian motion, so

$$dp(t, T) = p(t, T) v(t, T) d\tilde{\beta}_t + p(t, T) v(t, T) (\theta_t - \langle n_t, \psi_t \rangle) dt.$$

Step 3 (finite-variation parts). Also $dp(\cdot, T) = \tilde{B} dN^T + N^T d\tilde{B}$, because $\tilde{B}$ is continuous and of finite variation. By uniqueness of the canonical decomposition of a continuous $\tilde{Q}$ -semimartingale,

$$d\tilde{B}_t = \tilde{B}_t v(t, T) (\theta_t - \langle n_t, \psi_t \rangle) dt \quad \text{on } [0, T], \text{ for every } T \in (0, T^*].$$

Step 4. For $T_1 < T_2$, subtracting the two identities on $[0, T_1)$ gives $(v(t, T_1) - v(t, T_2))(\theta_t - \langle n_t, \psi_t \rangle) = 0$, $dt \otimes d\mathbb{P}$ -a.e. The function $y \mapsto y/(e^y - 1)$ is strictly decreasing on $(0, \infty)$ and $X_t > 0$, so $v(t, T_1) < v(t, T_2)$; hence $\theta = \langle n, \psi \rangle$ a.e. on $[0, T_1)$, for every $T_1 < T^*$. By Step 3, $d\tilde{B} = 0$ on $[0, T^*)$, so $\tilde{B} \equiv 1$ on $[0, T^*]$ by continuity, and $p(0, T^*) = E^{\tilde{Q}}[\tilde{B}_0/\tilde{B}_{T^*}] = 1$. This contradicts $p(0, T^*) = 1 - e^{-x_0/(2T^*)} < 1$. $\square$

The argument identifies the only possible account, $\tilde{B} \equiv 1$, and the only possible market price of risk along β , namely θ , for which $\mathcal{E}(-\int \theta d\beta) = \zeta$ is a strict local martingale; the contradiction is reached without using the latter fact.

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